

Unit 107 - Field Data Review

Mohawk Valley West Slope demonstration. Method **Tractor**, 95.3 ac, BAF 20 variable-radius cruise on a 300 ft grid, 46 plots, 522 trees tallied. Simulated data with planted recording errors; see docs/methods.md.

Table 1. Stand summary from the BAF 20 cruise, per acre

STAND DENSITY		STAND INFO	
Basal area	227 sq ft/ac (SE 1.6 %)	Type	Sierran mixed conifer
Trees per acre	72	CWHR size / density	4 / D (cover 93 %)
QMD	24.0 in	SDI	280 (48 % of max 584)
SAWTIMBER (>= 10 in)		BIOMASS (< 10 in)	
BA QMD	226 sq ft/ac 24.4 in	BA QMD	1 sq ft/ac 8.9 in
TPA	69	TPA	3
Volume	53.1 MBF/ac (9648 cu ft)	Volume	0.9 green tons/ac (30 cu ft)
TREATMENT TARGET		CRUISE DESIGN	
Leave	35 % of max SDI = ~165 sq ft/ac BA	Sampling error	3.3 % at 95 % (t = 2.01, n = 46)
Species (BA)	WF 71, PP 65, IC 37, DF 33, SP 21	Stratum standard	40 % - MEETS; 46 plots vs 20-plot practice: ok

Cubic volume by form factor 0.42 net of defect; Scribner board feet at 5.5 bf/cu ft (Keegan et al. 2010); green tons at 60 lb/cu ft. SDI in the summation form; maximum is the basal-area-weighted FVS Western Sierra species maximum. Sampling error at 95 % confidence with Student's t; stratum standard 40 % for tree-measurement sales (FSH 2409.12 ch. 40, 41.1).

Table 2. Stand table by 4-inch DBH class, per acre

DBH class, in	Trees	TPA	BA	cu ft	MBF
8-11.9	7	5.6	3.0	76	0.4
12-15.9	33	13.0	14.3	401	2.2
16-19.9	62	15.5	27.0	886	4.9
20-23.9	79	13.5	34.3	1286	7.1
24-27.9	82	9.9	35.7	1439	7.9
28-31.9	61	5.5	26.5	1100	6.0
32-35.9	54	3.8	23.5	1071	5.9
36-39.9	42	2.4	18.3	891	4.9
40-43.9	30	1.4	13.0	676	3.7
44-47.9	18	0.7	7.8	429	2.4
48-51.9	18	0.6	7.8	443	2.4
52-55.9	14	0.4	6.1	374	2.1
56-59.9	8	0.2	3.5	204	1.1
60-63.9	14	0.3	6.1	401	2.2
All	522	72.5	227.0	9677	53.2

Table 3. Stock table by species, per acre

Species	Trees	TPA	BA	QMD	cu ft	MBF
WF	164	22.5	71.3	24.1	3063	16.8
PP	149	20.1	64.8	24.3	2729	15.0
IC	85	13.1	37.0	22.8	1576	8.7
DF	76	10.2	33.0	24.4	1383	7.6
SP	48	6.5	20.9	24.2	926	5.1
All	522	72.5	227.0	24.0	9677	53.2

Trees = trees tallied in the cruise; TPA, BA (sq ft/ac), cubic feet and Scribner MBF per acre by BAF expansion. Species: PP ponderosa pine, WF white fir, DF Douglas-fir, SP sugar pine, IC incense-cedar.

Unit 107: plots on the 300 ft grid (orange = QA flag)

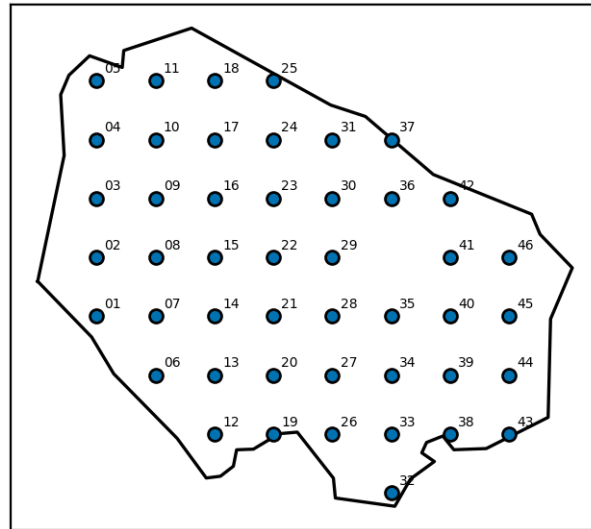


Figure 1. Plot locations with QA status.

QA findings: no issues found; unit accepted

Table 4. Plot list

Plot	Trees in	BA sq ft/ac	Slope %	Status
107-01	13	260	28	ok
107-02	12	240	18	ok
107-03	13	260	110	ok
107-04	11	220	31	ok
107-05	8	160	62	ok
107-06	11	220	20	ok
107-07	11	220	43	ok
107-08	13	260	7	ok
107-09	12	240	55	ok
107-10	10	200	56	ok
107-11	12	240	61	ok
107-12	11	220	52	ok
107-13	10	200	64	ok
107-14	13	260	127	ok
107-15	13	260	59	ok
107-16	11	220	55	ok
107-17	9	180	125	ok
107-18	12	240	47	ok
107-19	12	240	88	ok
107-20	10	200	87	ok
107-21	12	240	60	ok
107-22	12	240	50	ok
107-23	11	220	103	ok
107-24	12	240	28	ok
107-25	10	200	97	ok
107-26	8	160	84	ok
107-27	10	200	50	ok
107-28	11	220	60	ok
107-29	12	240	149	ok
107-30	11	220	32	ok
107-31	11	220	66	ok
107-32	12	240	78	ok
107-33	10	200	93	ok
107-34	11	220	24	ok
107-35	11	220	58	ok
107-36	11	220	24	ok
107-37	12	240	67	ok
107-38	12	240	108	ok
107-39	12	240	103	ok
107-40	12	240	61	ok
107-41	12	240	59	ok
107-42	13	260	107	ok
107-43	14	280	68	ok
107-44	10	200	76	ok
107-45	11	220	46	ok
107-46	12	240	60	ok